Answers: Logic Drill #4

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A

1. Give a counterexample to: "Every prime number is odd."

Answer: 2

Method: 2 is a prime number because its only positive divisors are 1 and 2, but 2 is even. A single counterexample disproves the universal claim.

> **Investigate further:** Are there any other even prime numbers?

2. Give a counterexample to: "$n^2 \geq n$ for every real number n."

Answer: $n = 0.5$ (or any $n \in (0, 1)$)

Method: For $n = 0.5 = 1/2$, we have $n^2 = 0.25$. Since $0.25 < 0.5$, $n^2 < n$, disproving the claim.

> **Investigate further:** For which interval of real numbers n is $n^2 < n$ true?

3. Give a counterexample to: "If n is prime, then $n + 2$ is prime."

Answer: $n = 7$ (or $n = 23$)

Method: 7 is prime, but $7 + 2 = 9 = 3^2$, which is composite.

> **Investigate further:** Primes p where $p+2$ is also prime are called twin primes. What is the smallest pair of twin primes?

4. Give a counterexample to: "Every multiple of 4 is a multiple of 8."

Answer: 4 (or $12, 20, 28, \ldots$)

Method: 4 is a multiple of 4 ($4 = 4 \times 1$), but 4 is not a multiple of 8 because $4/8 = 0.5 \notin \mathbb{Z}$.

> **Investigate further:** Is every multiple of 8 a multiple of 4?

5. To prove "there is no largest even integer" by contradiction, what should you assume first?

Answer: Assume that there exists a largest even integer (call it E).

Method: A proof by contradiction begins by assuming the exact negation of the statement to be proved.

> **Investigate further:** How do you then construct an even integer strictly larger than E?

6. True or false: "If assuming 'not P' leads to something impossible, P must be true."

Answer: True

Method: This is the fundamental principle of proof by contradiction (*reductio ad absurdum*). Since a statement cannot be both true and false, establishing that $\neg P$ leads to a contradiction forces P to be true.

> **Investigate further:** How does proof by contradiction differ from proof by contrapositive?

7. Give a counterexample to: "$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$."

Answer: $a = 1, b = 1$

Method: For $a = 1, b = 1$, $\sqrt{a+b} = \sqrt{2}$, whereas $\sqrt{a} + \sqrt{b} = 1 + 1 = 2$. Since $\sqrt{2} \neq 2$, the identity fails.

> **Investigate further:** Under what condition on a and b does $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ hold?

8. Give a counterexample to: "$(x+y)^2 = x^2 + y^2$ for all real x, y."

Answer: $x = 1, y = 1$

Method: For $x = 1, y = 1$, $(x+y)^2 = 2^2 = 4$, but $x^2 + y^2 = 1^2 + 1^2 = 2$. Since $4 \neq 2$, the claim is false.

> **Investigate further:** Expand $(x+y)^2$ to state the exact condition required for $(x+y)^2 = x^2 + y^2$.

9. True or false: "A single counterexample is enough to disprove a 'for all' claim."

Answer: True

Method: The negation of $\forall x, P(x)$ is $\exists x, \neg P(x)$. Finding one single element x where $P(x)$ is false satisfies the negation, disproving the universal claim.

> **Investigate further:** Is a single example enough to prove a 'for all' claim?

10. Give a counterexample to: "$n! + 1$ is always prime."

Answer: $n = 4$ (or $n = 5$)

Method: For $n = 4$, $4! + 1 = 24 + 1 = 25 = 5^2$, which is composite. (For $n = 5$, $5! + 1 = 121 = 11^2$).

> **Investigate further:** Is $n! + 1$ ever prime for $n > 3$?

Section B

1. Five envelopes contain the numbers $6, 9, 14, 21, 25$ respectively. Each is sealed with a wax stamp showing either a star or a circle: the envelopes containing $6, 14, 25$ are star-sealed, and the envelopes containing $9, 21$ are circle-sealed. Claim: "Every star-sealed envelope contains an even number." Which envelope's contents provide a counterexample?

Answer: 25

Method: The star-sealed envelopes contain $6, 14, 25$. 6 and 14 are even, but 25 is odd. The envelope containing 25 is star-sealed yet contains an odd number, disproving the claim.

> **Investigate further:** How does this relate to Wason's Selection Task in logic?

2. Give a counterexample to: "If a and b are both irrational, then $a + b$ is irrational."

Answer: $a = \sqrt{2}, b = -\sqrt{2}$

Method: $\sqrt{2}$ and $-\sqrt{2}$ are both irrational numbers, but their sum is $\sqrt{2} + (-\sqrt{2}) = 0 = 0/1 \in \mathbb{Q}$, which is rational.

> **Investigate further:** Can you construct a counterexample where both a and b are positive?

3. Give a counterexample to: "If a and b are both irrational, then ab is irrational."

Answer: $a = \sqrt{2}, b = \sqrt{2}$

Method: $\sqrt{2}$ is irrational, but $ab = \sqrt{2} \times \sqrt{2} = 2 = 2/1 \in \mathbb{Q}$, which is rational.

> **Investigate further:** Find a counterexample where $a \neq b$.

4. Prove by contradiction: "There is no largest even integer."

Answer: Proof by contradiction

Method: Assume toward a contradiction that there exists a largest even integer, E. By definition of even integers, $E = 2k$ for some integer k. Consider $E' = E + 2 = 2(k + 1)$. Since $k + 1 \in \mathbb{Z}$, E' is an even integer. Furthermore, $E' > E$. This contradicts the assumption that E was the largest even integer. Thus, no largest even integer exists.

> **Investigate further:** Why must we add 2 rather than 1 to maintain even parity?

5. Give a counterexample (using $n = 0$) to: "For every integer n, $n/n = 1$."

Answer: $n = 0$

Method: For $n = 0$, $0/0$ is undefined in standard arithmetic. Since $0/0 \neq 1$, $n = 0$ is a counterexample.

> **Investigate further:** What happens to the expression x/x as $x \to 0$ in calculus?

6. Give a counterexample (using $n = 1$) to: "Every positive integer has at least one prime factor."

Answer: $n = 1$

Method: The prime factorisation of 1 is empty because 1 has no prime factors (it is neither prime nor composite). Thus $n = 1$ is a counterexample.

> **Investigate further:** Why is 1 excluded from being a prime number in the Fundamental Theorem of Arithmetic?

7. Prove by contradiction: "There is no smallest positive rational number."

Answer: Proof by contradiction

Method: Assume toward a contradiction that $r \in \mathbb{Q}^+$ is the smallest positive rational number. Since $r \in \mathbb{Q}^+$, $r = a/b$ for positive integers a, b. Consider $q = r/2 = a/(2b)$. Since $a, 2b$ are positive integers, $q \in \mathbb{Q}^+$. Furthermore, since $r > 0$, we have $0 < q < r$. This contradicts r being the smallest positive rational number. Hence, no smallest positive rational number exists.

> **Investigate further:** Does this property (density) hold for integers?

8. Give a counterexample to: "If p and q are distinct primes, then $p + q$ is composite."

Answer: $p = 2, q = 3$

Method: 2 and 3 are distinct primes, but $p + q = 2 + 3 = 5$, which is prime (not composite).

> **Investigate further:** If p and q are both odd distinct primes, is $p + q$ always composite?

9. Give a counterexample to: "For every positive integer $n \geq 1$, $2^n - 1$ is prime."

Answer: $n = 4$ (or $n = 1, n = 6$)

Method: For $n = 4$, $2^4 - 1 = 15 = 3 \times 5$, which is composite.

> **Investigate further:** Primes of the form $2^p - 1$ are called Mersenne primes. Must n be prime for $2^n - 1$ to be prime?

10. Prove by contradiction: "If n is a positive integer, n and $n + 1$ share no common factor greater than 1."

Answer: Proof by contradiction

Method: Suppose toward a contradiction that there exists a common factor $d > 1$ of n and $n + 1$. Then $d \mid n$ and $d \mid (n + 1)$. Therefore, $d \mid ((n + 1) - n) \implies d \mid 1$. But the only positive divisor of 1 is 1, which contradicts $d > 1$. Thus, $\gcd(n, n + 1) = 1$.

> **Investigate further:** Generalise this: What is the maximum possible common factor of n and $n + k$?

Section C

1. Consider the statement: "($*$) If a whole number n is 2 more or 4 more than a multiple of 5, then n is prime." How many counterexamples to ($*$) are there in the range $1 \leq n \leq 50$?

Answer: 13

Method: *(after TMUA 2019 Paper 2 Q4)* Numbers in $1 \leq n \leq 50$ of the form $5k+2$ are: $2, 7, 12, 17, 22, 27, 32, 37, 42, 47$. Of these, the composites are $12, 22, 27, 32, 42$ (5 counterexamples).
Numbers of the form $5k+4$ are: $4, 9, 14, 19, 24, 29, 34, 39, 44, 49$. Of these, the composites are $4, 9, 14, 24, 34, 39, 44, 49$ (8 counterexamples).
Total counterexamples $= 5 + 8 = 13$.

> **Investigate further:** Why is 2 not a counterexample despite being $5(0) + 2$?

2. Find the smallest positive integer n for which $n^2 + n + 41$ is *not* prime.

Answer: 40

Method: For $n = 40$, $n^2+n+41 = 40^2+40+41 = 40(40+1)+41 = 40(41)+41 = 41(40+1) = 41^2 = 1681$, which is composite. For all $1 \leq n \leq 39$, $n^2 + n + 41$ is prime.

> **Investigate further:** Is $n = 41$ also a counterexample?

3. Give a counterexample to: "$n^2 + 1$ is always either prime or a perfect square, for every positive integer n."

Answer: $n = 3$ (or $n = 7, 8$)

Method: For $n = 3$, $n^2 + 1 = 3^2 + 1 = 10$. 10 is not prime ($10 = 2 \times 5$) and 10 is not a perfect square ($3^2 < 10 < 4^2$). Thus $n = 3$ disproves the claim.

> **Investigate further:** For $n = 7$, what is $n^2 + 1$?

4. Prove by contradiction: "There is no integer solution to $x^2 - 3y^2 = 2$."

Answer: Proof by contradiction

Method: Suppose toward a contradiction that there exist integers x, y such that $x^2 - 3y^2 = 2$. Taking this equation modulo 3, we get $x^2 \equiv 2 \pmod 3$. However, for any integer x, $x \equiv 0, 1$, or $2 \pmod 3$, which gives $x^2 \equiv 0, 1$, or $1 \pmod 3$. Thus $x^2 \pmod 3 \in \{0, 1\}$, so $x^2 \equiv 2 \pmod 3$ has no solutions. Contradiction.

> **Investigate further:** Which other integer remainders cannot be quadratic residues?

5. Find the smallest positive integer n for which $3^n + 2$ is *not* prime.

Answer: 5

Method: Testing $n = 1, 2, 3, 4, 5$:
$n = 1 \implies 3^1 + 2 = 5$ (prime);
$n = 2 \implies 3^2 + 2 = 11$ (prime);
$n = 3 \implies 3^3 + 2 = 29$ (prime);
$n = 4 \implies 3^4 + 2 = 83$ (prime);
$n = 5 \implies 3^5 + 2 = 243 + 2 = 245 = 5 \times 49$ (composite). Smallest n is 5.

> **Investigate further:** Why is $3^n + 2$ always divisible by 5 when n is an odd multiple of 5?

6. Give a counterexample to: "If x and y are both non-integers, then $x + y$ is a non-integer."

Answer: $x = 0.5, y = 0.5$

Method: $x = 0.5 \notin \mathbb{Z}$ and $y = 0.5 \notin \mathbb{Z}$, but $x + y = 1.0 = 1 \in \mathbb{Z}$, which is an integer.

> **Investigate further:** Find a counterexample where x and y are not rational numbers.

7. Prove by contradiction: "If a, b are positive integers and $a^2 + b^2$ is odd, then a and b do not have the same parity."

Answer: Proof by contradiction

Method: Suppose toward a contradiction that a and b have the same parity.
Case 1: Both a, b are even. Then $a = 2k, b = 2m \implies a^2 + b^2 = 4k^2 + 4m^2 = 2(2k^2 + 2m^2)$, which is even.
Case 2: Both a, b are odd. Then $a = 2k+1, b = 2m+1 \implies a^2 + b^2 = (4k^2 + 4k + 1) + (4m^2 + 4m + 1) = 2(2k^2 + 2k + 2m^2 + 2m + 1)$, which is even.
In both cases $a^2 + b^2$ is even, contradicting $a^2 + b^2$ being odd. Thus a and b must have opposite parity.

> **Investigate further:** What can you deduce about $a^2 + b^2 \pmod 4$ when a and b have opposite parity?

8. Give a counterexample to: "$n^2 - n + 17$ is prime for every positive integer n."

Answer: $n = 17$ (or $n = 18$)

Method: For $n = 17$, $n^2 - n + 17 = 17^2 - 17 + 17 = 17^2 = 289$, which is composite ($17 \times 17$).

> **Investigate further:** What is the value of $n^2 - n + 17$ for $n = 18$?

Section D

1. Consider: I. "$n = 41$ is a counterexample to '$n^2 + n + 41$ is always prime'." II. "$n = 40$ is the smallest counterexample to '$n^2 + n + 41$ is always prime'." III. "$n = 4$ is a counterexample to '$n^2 + n + 41$ is always prime'." Which of I, II, III are true?

Answer: E) I and II only

Method: I: $41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \times 43$, composite. So I is true.
II: For $1 \leq n \leq 39$, $n^2 + n + 41$ is prime. At $n = 40$, $40^2 + 40 + 41 = 41^2$, composite. Thus $n = 40$ is indeed the smallest positive integer counterexample. II is true.
III: For $n = 4$, $4^2 + 4 + 41 = 61$, which is prime. So $n = 4$ is not a counterexample. III is false.
Hence, I and II only are true.

> **Investigate further:** Why does Euler's polynomial $n^2 + n + 41$ yield primes for all n from 0 to 39?

2. Prove by contradiction: "There do not exist positive integers a, b with $a^2 - b^2 = 1$."

Answer: Proof by contradiction

Method: Assume toward a contradiction that there exist positive integers a, b such that $a^2 - b^2 = 1$. Factorising as a difference of two squares gives $(a - b)(a + b) = 1$. Since $a, b \in \mathbb{Z}^+$, we have $a \geq 1$ and $b \geq 1$, so $a + b \geq 2$. However, the only factorisation of 1 into positive integers is 1×1, requiring $a + b = 1$. This contradicts $a + b \geq 2$. Thus, no such positive integers exist.

> **Investigate further:** Do non-zero integer solutions exist if a, b are not restricted to positive integers?

3. A student claims: "For every positive integer n, at least one of $n, n + 2, n + 4$ is divisible by 3." Determine, with proof, whether this claim is true.

Answer: True

Method: Proof: Any positive integer n can be written as $3k$, $3k + 1$, or $3k + 2$ for some integer $k \geq 0$.
Case 1: If $n = 3k$, then n is divisible by 3.
Case 2: If $n = 3k + 1$, then $n + 2 = (3k + 1) + 2 = 3k + 3 = 3(k + 1)$, which is divisible by 3.
Case 3: If $n = 3k + 2$, then $n + 4 = (3k + 2) + 4 = 3k + 6 = 3(k + 2)$, which is divisible by 3.
In all cases, at least one of $n, n + 2, n + 4$ is divisible by 3. The claim is true.

> **Investigate further:** Can more than one of $n, n + 2, n + 4$ be divisible by 3 for the same n?

4. Give a counterexample to: "If p is prime, then $2^p + 1$ is prime."

Answer: $p = 3$ (or $p = 5$)

Method: For $p = 3$ (which is prime), $2^p + 1 = 2^3 + 1 = 8 + 1 = 9 = 3^2$, which is composite. (Also for $p = 5$, $2^5 + 1 = 33 = 3 \times 11$).

> **Investigate further:** For which prime p is $2^p + 1$ actually prime?

5. Statement $(*)$: "For every positive integer $n \geq 2$, there is a prime strictly between n and $2n$." (a) Verify $(*)$ for $n = 2, 3, 4, 5$ by direct computation. (b) Does this verification constitute a proof of $(*)$? Justify your answer.

Answer: (a) $n = 2 : 3$; $n = 3 : 5$; $n = 4 : 5$ or 7; $n = 5 : 7$. (b) No, finite checking does not prove a universal statement.

Method: (a) For $n = 2$, prime $3 \in (2, 4)$. For $n = 3$, prime $5 \in (3, 6)$. For $n = 4$, prime $5 \in (4, 8)$. For $n = 5$, prime $7 \in (5, 10)$.
(b) No. Verification for finitely many cases ($n = 2, 3, 4, 5$) does not constitute a proof for infinitely many positive integers $n \geq 2$. A single unchecked value could be a counterexample. (The statement is true for all $n \geq 2$, known as Bertrand's Postulate, but proving it requires a general argument).

> **Investigate further:** What is the formal name of the theorem stating that a prime always exists between n and $2n$?

"One counterexample kills a universal claim. Four confirmations do nothing whatsoever to save it."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths